FLUOR DANIEL

COMBINED FOUNDATIONS

PURPOSE

This practice establishes guidelines and recommended procedures to size and design a combined foundation or mat supported on soil.

SCOPE

This practice includes the following major sections:

- DEFINITIONS
- DESIGN CONDITIONS
- PEDESTAL DESIGN
- FOUNDATION SIZING
- SOIL BEARING
- FOUNDATION DESIGN
- REFERENCES
- ATTACHMENTS

APPLICATION

This practice is to be used to design a single combined foundation or mat to carry all of the applied loads, whenever 2 or more pieces of equipment, building supports, or structure columns are so closely spaced that it is impractical or impossible to design individual foundations for each item.

DEFINITIONS

Combined Foundation: Any foundation supporting more than a single column or wall load.

Mat Foundation: A large concrete slab supporting a number of columns or pedestals.

DESIGN CONDITIONS

Applied Loads

The vertical and horizontal loads and overturning moments acting on the foundation will be the sum of all the vertical and horizontal loads and overturning moments from each individual item supported by the foundation. The loads from each item shall be determined in accordance with the procedure provided in the practice for that particular item.

Load Combinations

All the load combinations specified in the job specifications shall be considered in determining the design forces (moments and shears) as well as the soil bearing pressures for the foundation.

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- One of the following load combinations will usually govern the design:
- Empty + wind or seismic
- Operating + wind or seismic
- Test (hydrotest)

Consider load reversal for seismic and wind loads in each of the principal areas of the foundation. In some cases, combining orthogonal seismic loads may be required. Refer to UBC (Uniform Building Code) Section 1631.1.

Critical maximum moment and shear may not necessarily occur with the largest simultaneously applied load at each column or pedestal.

When hydrotest is a design consideration, the Design Engineer will verify whether the hydrotest will involve only 1 vessel at a time or if it is possible that more than 1 vessel can be simultaneously hydrotested. If more than 1 vessel supported on the same foundation is to be simultaneously hydrotested, the sequence of hydrotesting will be investigated and be taken into consideration.

Design Loads

Design loads (moments and shears) within the foundation will be determined by analysis using the appropriate load factors specified by ACI (American Concrete Institute) 318.

The design loads will not exceed the capacity of the concrete section as determined by the provisions of ACI 318.

The maximum applied soil bearing pressure will be computed using unfactored (working) loads and will not exceed the allowable soil bearing pressure recommended by the Geotechnical Engineer for that foundation.

PEDESTAL DESIGN

The various pedestals and columns supported by the foundation, as well as the anchor bolts attaching the structures or equipment to the pedestals or columns, will be designed in accordance with the procedures provided in the practice for each particular item.

FOUNDATION SIZING

Center Of Gravity (CG)

The centroid of the contact area between the soil and the concrete foundation or mat will be set to coincide with the overall CG of all the sustained (long-term) loads, including the self-weight of the combined foundation or mat. This requirement, in combination with the arrangement of the various items being supported, will determine the minimum size and shape of the foundation or mat.

In some instances, it may not be possible or practical to align the centroid of the foundation contact area with the CG of the sustained loads. In such a case, use engineering judgment and account for the effects of the resulting permanent eccentricity in the analysis and design of the foundation. Generally, due to the potential for differential settlements, a permanent eccentricity greater than 5 percent of the respective foundation dimension is considered to be unacceptable.

Thickness

The minimum foundation thickness will be as follows:

- 12 inches.
- Minimum thickness to develop the pedestal reinforcement.
- Minimum thickness required to resist 2-way (punching) shear without reinforcement at each pedestal or column.
- Minimum thickness to resist 1-way (diagonal tension) shear without reinforcement.
- Minimum thickness to ensure rigidity for manual analysis.

Note!!! Whenever evaluating 2-way shear requirements, consider the effects of the applied moment at the base of the pedestal or column in addition to the effects of the applied vertical load; similar to the way it would be considered for slabs in ACI 318, Section 11.12.2.

Depth

The depth of the foundation below finished grade is usually given in the job specifications based on recommendations from the Geotechnical Engineer. However, the Design Engineer will verify that the following conditions are met:

- Bottom of foundation is located below the frost level.
- Adequate clearance is provided between the top of the foundation and finished grade to allow placement of underground piping and electrical.
- Sufficient embedment is provided to ensure overall stability.

Stability Ratio

The minimum stability ratio (minimum resisting moment/overturning moment) for all unfactored load combinations will be as specified in the job specifications.

SOIL BEARING

Except for extremely unusual problems, assume the soil bearing pressure at the base of a combined foundation to follow either a distribution governed by elastic subgrade reaction (nonlinear) or a straight line (linear) distribution. In either case, the resultant of the soil bearing pressure will coincide in location, and will be equal and of opposite direction to the resultant of the applied loads. At no place will the calculated soil bearing pressure exceed the maximum allowable soil bearing pressure.

For large combined foundations and mats, the allowable soil bearing pressure will be determined with consideration made for potential differential settlements with respect to adjacent structures or foundations.

The allowable soil bearing pressure will not be increased for any load combination unless specifically allowed by the job specification.

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Linear Distribution

Assume a linear soil bearing pressure distribution for all foundations that can be considered rigid; that is to say, very small deformations will result from the applied loading. Criteria as to whether or not a foundation is rigid are given below.

Assume a linear soil bearing pressure distribution in cases where the footing design is governed by shear stresses.

Whenever 1 item, such as a tower, contributes virtually all the load and moment affecting the foundation, the foundation is normally considered rigid, and a linear distribution of soil pressure is assumed.

If 2 or more items such as 2 vertical vessels contribute approximately equal loads and moments to the foundation, the engineer may assume a rigid foundation for the purpose of determining maximum allowable soil bearing pressure. However, use a strip analysis or similar procedure in determining the footing stresses.

Nonlinear Distribution

A nonlinear soil bearing pressure distribution, found by means of subgrade reaction theories, is based on a simplification of the complex relationship between the bearing pressures and the corresponding soil deformations. If applied to subsoils of such character that the deformations are localized in the general vicinity of the loads, and if the maximum bearing pressures are smaller than about 1/2 the ultimate bearing capacity of the soil, this analysis provides reasonably accurate results.

The modulus of subgrade reaction depends on many variables and should be provided by the Geotechnical Engineer in the site specific soil report. Refer to Attachment 01, Table 1, for a range of typical values for use in preliminary designs and estimates.

Refer to Bowles' "Foundation Analysis and Design," for a general discussion on the modulus of subgrade reaction and the effects of foundation shape, size, and depth.

FOUNDATION DESIGN

Rigidity Check

The following criteria to determine if a foundation is rigid are based on ACI 336.2R, "Suggested Analysis and Design Procedures for Combined Footings and Mats", and "Foundation Engineering Handbook".

A foundation will be considered rigid if the relative stiffness factor, K_r, is greater than or equal to 0.5.

The relative stiffness factor, K_r, compares the overall stiffness of the foundation, structural framing members, and shear walls with that of the underlying soil.

To determine the relative stiffness factor, make an analysis comparing the combined stiffness of the foundation, superstructure framing members, and shear walls with the stiffness of the soil.

That is,

$$K_r = \frac{E_B I}{E_s' b^3}$$ **Equation 1**

and

$$E_B I_B = \left[E' I_F + \sum E' I_b' + \sum \frac{E' a h^3}{12}\right] / b$$ **Equation 2**

where

E'	= Modulus of elasticity of the materials used in the respective portions of the structure and foundation (ksf)
I'_b	= Moment of inertia of the structural framing members (ft4)
I_F	= Moment of inertia of the foundation (ft4)
E'_s	= Modulus of elasticity of the soil (ksf)
a	= Thickness of shear wall (ft)
h	= Height of shear wall (ft)
b	= Width of footing (ft)

It would be conservative to ignore the stiffness of the structural framing members and shear walls when computing K_r.

The modulus of elasticity of the soil, E'_s, should be provided by the Geotechnical Engineer in the site specific soil report. Refer to Attachment 01, Table 2 for a typical range of values for use in preliminary designs and estimates.

Alternatively, a foundation or mat will be considered rigid if the pedestal or column loads and spacings are relatively uniform (vary by no more than 20 percent) and the pedestal or column spacing is less than 1.75/λ.

$$\lambda = \left(\frac{k_b b}{4 E_c I}\right)^{1/4}$$

where

k_b	= Modulus of subgrade reaction (kcf)
b	= Width of a strip of foundation or mat between centers of adjacent bays (ft)
E_c	= Modulus of elasticity of concrete (ksi)
I	= Moment of inertia of a foundation strip of width b (ft4)

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Analysis And Design

Rigid Foundations

Analyze foundations and mats that have been determined to be rigid on the basis of simple static and rigid body mechanics.

Once the soil bearing has been determined, develop a shear and moment diagram for each principal direction of the foundation.

Take the critical sections for moment and shear with respect to the face of a square of area equivalent to that of the major pedestal. If there is not a major pedestal, take the critical sections for moment and shear at the centerline of the pedestal.

Determine the external moment on any section by passing through the section a vertical plane that extends completely across the footing, and compute the moment of the forces acting over the entire area of the footing on one side of the plane.

Check shear as a measure of diagonal tension according to ACI 318.

Place the bottom reinforcing required at the critical sections continuously across the entire footing. If possible, avoid variations in reinforcing sizes for opposite directions.

Top steel, as required for concentrated loads and uplift, may or may not be distributed across the entire footing width depending on each individual case.

Strip Analysis

Analyze rigid foundations with more than 1 major load source using a strip analysis as outlined below.

Strips should be taken across the entire footing in the area of the major loads. The strip width will normally be the width of the equivalent square pedestal and 3 times the footing thickness, or 1- 1/2 the footing thickness on each side of the equivalent square.

Next, determine the soil bearing pressures required to produce equilibrium with all the loads and moments acting on the strip.

Determine critical shear and moment on the basis of simple statics, and place the resulting reinforcing continuously across the entire footing.

Flexible Foundations

The analysis and design of a nonrigid (flexible) foundation or mat is very rigorous and time consuming. A description of various methods such as beam or elastic foundation can be found in soils and foundation textbooks. Because of the difficulty and time involved, computerized methods should be used for the analysis of flexible foundations or mats.

Once the foundation has been accurately modeled and analyzed, the location and magnitude of the critical moments and shears can be determined. From this, the required foundation thickness and reinforcement can be evaluated.

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Computer Analysis

Rigid Foundations

A 2 dimensional computer structural analysis program can be used to analyze a rigid foundation or mat.

The applied loading and resulting soil bearing pressure will be determined for each load combination to be considered. The foundation will be modeled with beam elements and all nodes except 1 are free to rotate and translate as controlled by the stiffness of the beam elements.

Overall stability is provided by fixing a node at 1 end of the foundation against any translations and rotations. Resulting forces and moments at this node provide a measure of the imbalance between the applied loads and the calculated soil bearing pressure used in the analysis.

The magnitude of these unbalanced forces and moments should be small in comparison to the moments and shears used for the design of the foundation.

The moments and shears determined by the analysis are used to develop the thickness and reinforcing requirements discussed above.

Flexible Foundations

Before starting a computerized analysis of a flexible foundation or mat, the Design Engineer will investigate which computer programs are available.

There are a great number of variables and uncertainties which must be addressed when creating a model for a computerized analysis of a flexible foundation or mat. Therefore, attention to details and good engineering judgment must be exercised.

Use conservative assumptions and work closely with the Geotechnical Engineer to develop realistic values for the stiffness of the soil and do not rely on values taken from text books.

The Design Engineer will have thorough understanding of the model and how the software will interpret the model so that a realistic solution can be obtained.

REFERENCES

ACI 336.2R-88. Suggested Analysis and Design Procedures for Combined Footings and Mats. American Concrete Institute. Farmington Hills, Michigan. 1988

ACI 318-95. Building Code Requirements for Structural Concrete and Commentary. American Concrete Institute. Farmington Hills, Michigan. 1995

UBC 94. Uniform Building Code. International Conference of Building Officials. Whittier, CA. 1994

Joseph E. Bowles. Foundation Analysis and Design. McGraw-Hill Inc. New York, NY. Fourth Edition, 1988

Hsai-Yang Fang. Foundation Engineering Handbook. Van Nostrand Reinhold Co. New York, NY. Second Edition, 1991

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Structural Engineering Practice 670.215.1207:	Anchor Bolt Design Criteria
Structural Engineering Practice 670.215.1215:	Wind Load Calculation
Structural Engineering Practice 670.215.1216:	Earthquake Engineering

ATTACHMENTS

Attachment 01:
Range of Typical Soil Values

Attachment 02:
Sample Design of a Combined Foundation

RANGE OF TYPICAL SOIL VALUES

Table 1: Range of Values Of Modulus Of Subgrade Reaction (K_s), (reprinted from Bowles)

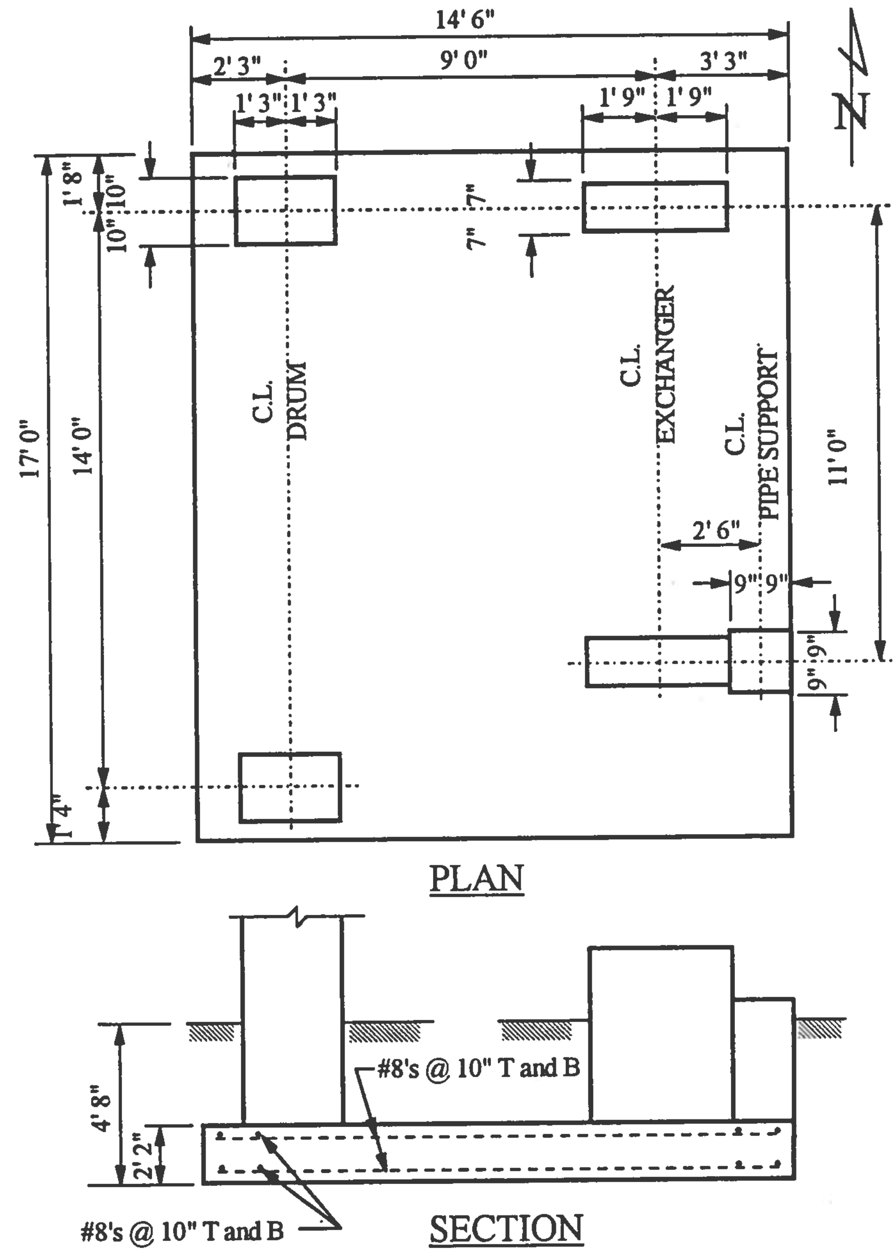
Soil	Range of k_s, kcf		
Loose sand	30	-	100
Medium sand	60	-	500
Dense sand	400	-	800
Clayey sand (medium)	200	-	500
Silty sand (medium)	150	-	300
Clayey soil:			
$q_u < 4$	75	-	150
$4 < q_u < 8$	150	-	300
$8 < q_u$		>	300

These values should be used for a guide. Local values may be higher or lower.

Table 2: Typical Range Of Values For Modulus Of Elasticity Of Soil (E'_s), (reprinted from Fang)

Type of Soil	Modulus of Elasticity, psi		
Very soft clay	50	-	400
Soft clay	250	-	600
Medium clay	600	-	1,200
Hard clay	1,000	-	2,500
Sandy clay	4,000	-	6,000
Silty sand	1,000	-	3,000
Loose sand	75	-	150
Dense sand	150	-	300
Dense sand and gravel		>	300

SAMPLE DESIGN OF A COMBINED FOUNDATION



Note: Drum and exchanger pier dimensions are the same at each end. Details and reinforcing are not shown. Slide plate details are not shown.

SAMPLE DESIGN OF A COMBINED FOUNDATION

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Design Data

Concrete:

28 Day Compression Strength, $f'_c = 4{,}000$ psi
ASTM 615 Grade 60 deformed bars, $f_y = 60{,}000$ psi
Reinforced concrete capacity determined per the provisions of ACI 318-89

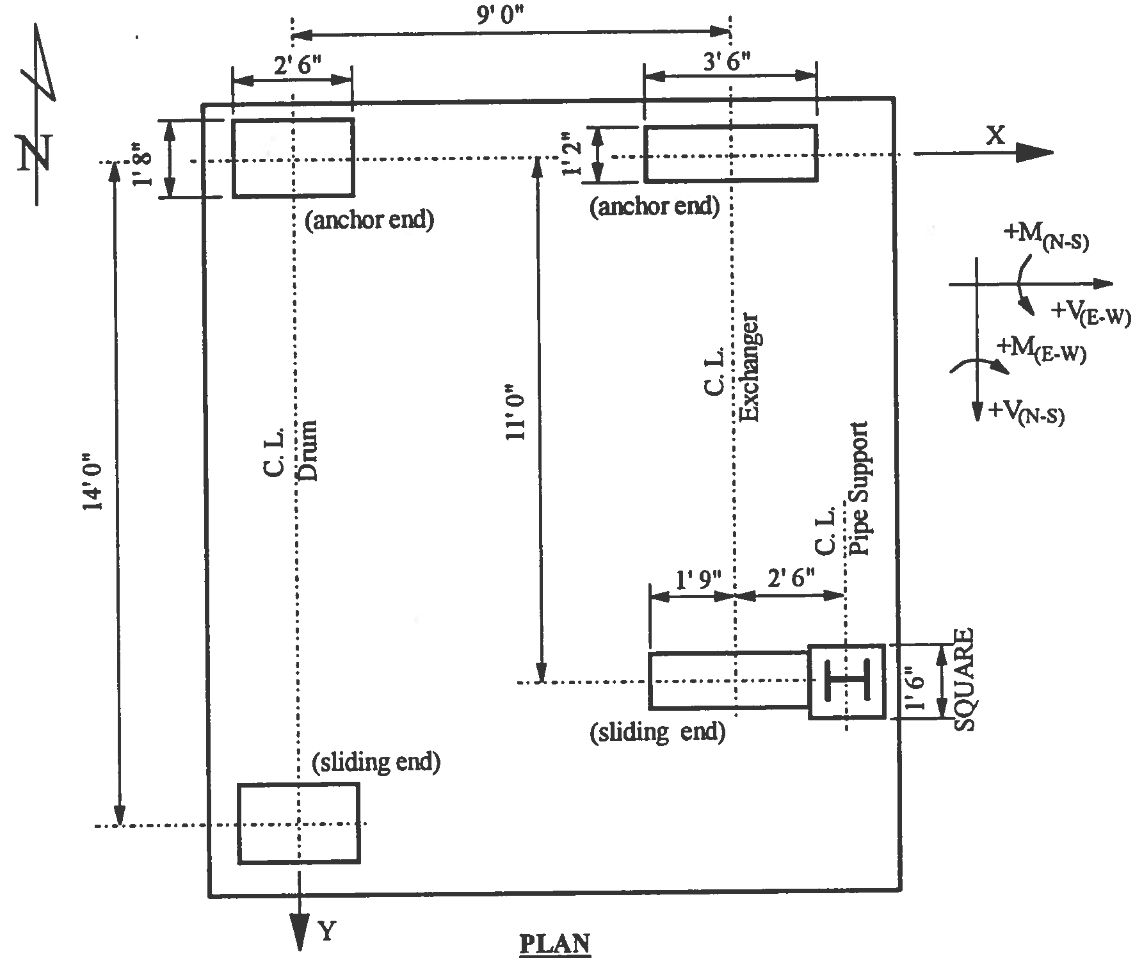
Soil: (medium clay)

Allowable soil bearing = 3,500 psf net (No 1/3 increase for seismic or wind)
Modulus of Elasticity, $E_s = 1{,}000$ psi
Modulus of Subgrade Reaction, $K_s = 100$ kcf
Soil Density, $\gamma = 100$ pcf
Coefficient of Friction Between foundation and soil, $\mu = 0.4$

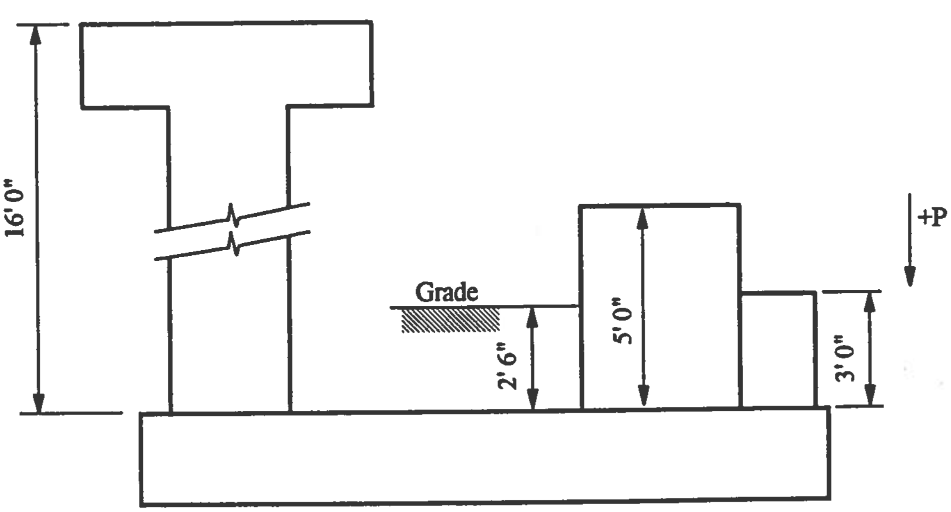
To allow placement of underground piping and electrical provide 2' 6" of earth cover over foundation.
The water table is well below the bottom of foundation.

SAMPLE DESIGN OF A COMBINED FOUNDATION

PIER LAYOUT



PLAN



ELEVATION

SAMPLE DESIGN OF A COMBINED FOUNDATION

SUMMARY OF LOADS APPLIED TO FOUNDATION MAT

		Empty			Operating			
		Eqpt & Pedestal	seismic N to S	seismic W to E	Eqpt & Pedestal	Friction (growth)	seismic N to S	seismic W to E
Drum (anchor end)	P	38 kips	-7.2 kips	-	60 kips	-	-13.7 kips	-
	V	-	22.9 kips	14.0 kips	-	-4.6 kips	39.1 kips	22.1 kips
	M	-	338.7 ft-k	246.3 ft-k		-73.6 ft-k	597.9 ft-k	421.7 ft-k
Drum (sliding end)	P	38 kips	7.2 kips	-	60 kips	-	13.7 kips	-
	V	-	5.2 kips	14.0 kips	-	4.6 kips	5.2 kips	22.1 kips
	M	-	55.5 ft-k	246.3 ft-k	-	73.6 ft-k	55.5 ft-k	421.7 ft-k
Exchanger (anchor end)	P	35.2 kips	-13.9 kips	-	46 kips	-	-18.9 kips	-
	V	-	30.2 kips	7.0 kips	-	-6.4 kips	40.5 kips	9.2 kips
	M	-	147.3 ft-k	66.0 kips	-	-32 ft-k	198.8 ft-k	89.8 ft-k
Exchanger (sliding end)	P	50.8 kips	13.9 kips	-	68 kips	-	18.9 kips	-
	V	-	0.8 kips	10.1 kips	-	6.4 kips	0.8 kips	13.5 kips
	M	-	2.0 ft-k	98.1 ft-k	-	32 ft-k	2.0 ft-k	133.2 ft-k
Pipe Support	P	2 kips	0	0	3 kips	-	0	0
	V	-	0.2 kips	0.2 kips	-	-	0.3 kips	0.3 kips
	M	1 ft-k	3 ft-k	3 ft-k	1.5 ft-k (E/W)	-	4.5 ft-k	4.5 ft-k

Notes:
1. Lateral and friction loads shown reflect possible load reversals. Positive load directions are defined on page ##.

2. Loads include weight of column or pier.

The details of exchanger pier (TP-1221), vessel pier (TP-1222), and seismic calculation (TP-1216) are ommitted for brevity. For information on this design, refer to practices 670.215.1221, 670.215.1222, and 670.215.1216 respectively.

In this example seismic loads control. If wind loading is a significant consideration, then such calculations must be included.

SIZING OF FOUNDATION/UNFACTORED LOADS

For center of gravity calculations, use centerline of drum at anchor end as origin as indicated on layout.

Determine Center of Gravity (Operating)

	P	Soil Displaced by Pier	P_{net}	North - South Direction		East - West Direction	
				Y	P_{net} Y	X	P_{net} X
Drum (anchor end)	60 kips	1 kip	59 kips	0	0	0	0
Drum (sliding end)	60 kips	1 kip	59 kips	14 ft	826 ft-k	0	0
Exchanger (anchor end)	46.4 kips	1 kip	45.4 kips	0	0	9 ft	409 ft-k
Exchanger (sliding end)	67.6 kips	1 kip	66.6 kips	11 ft	733 ft-k	9 ft	599 ft-k
Pipe Support	3 kips	0.6 kip	2.4 kips	11 ft	26 ft-k	11.5 ft	28 ft-k
Totals =			232.4 kips		1,585 ft-k		1,036 ft-k

$Y_{na} = \Sigma (P_{net}\ Y) / \Sigma (P_{net}) = (1{,}585 \text{ ft-k}) / (232.4 \text{ kips}) = 6.82 \text{ ft}$

$X_{na} = \Sigma (P_{net}\ X) / \Sigma (P_{net}) = (1{,}036 \text{ ft-k}) / (232.4 \text{ kips}) = 4.46 \text{ ft}$

Determine Center of Gravity (Empty)

Calculations are omitted for brevity. The results are:

$P_{net} = 159.4$ kips $Y_{na} = 6.78$ ft, $X_{na} = 4.84$ ft

Trial Mat Size and Location Based on the Operating C. G. (refer to sketch on next page)

Note: This sample calculation designs the mat as a rigid body in order to simplify the calculation. A thinner mat could be designed using a flexible mat analysis. A flexible mat analysis would normally require a finite element computer analysis.

Determine Mat Thickness Required for Rigid Behavoir

Concrete moment of inertia, (ACI 318, section 8.5.1)
$E_B = 57{,}000 \sqrt{f'_c} = 57{,}000 \sqrt{4{,}000 \text{ psi}} = 3{,}605{,}000$ psi, or 3,605 ksi

Soil modulus of elasticity (from design data), $E'_s = 1{,}000$ psi, or 1 ksi

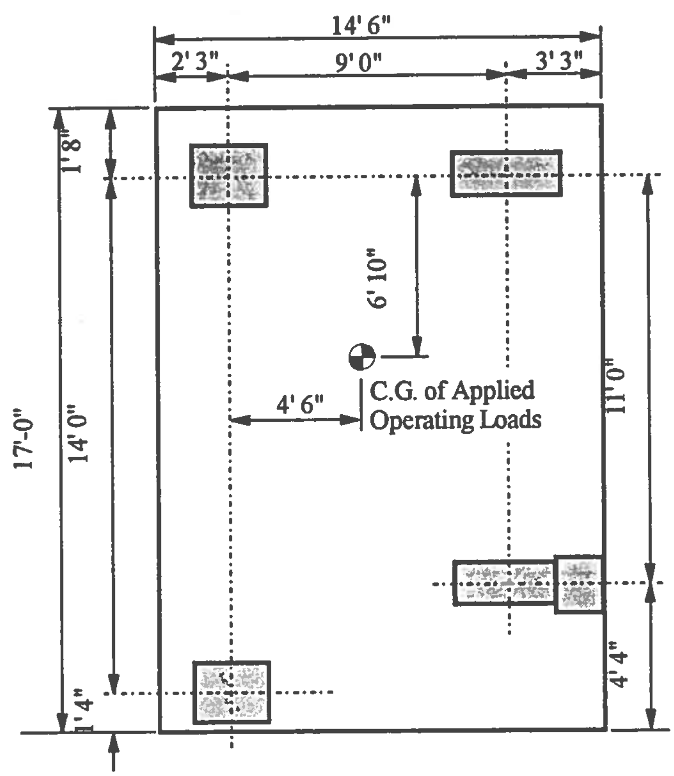
Length of mat, b = 17.0 ft

Unit moment of inertia of foundation ignoring stiffness of vessels (from Practice), $I_B = [b_f t^3 / 12] / b_f$

Stiffness criteria (from Practice), $K_r = E_B I_B / E'_S b^3 > 0.5$

Rearranging stiffness equation,
$I_B > [0.5 E'_S b^3] / E_B$

SAMPLE DESIGN OF A COMBINED FOUNDATION



Substituting for I_B
$t_{req} = [6\ E'_s\ b^3 / E_B]^{1/3} = [6\ (1\ ksi)(17\ ft)^3 / (3{,}605\ ksi)]^{1/3} = 2.01\ ft > 12$ in minimum

Pier reinforcing development should also be considered. <u>USE 2' 2" thickness</u>

Check Offset Criteria

By inspection of trial mat location, the east/west direction is not centered.

	density times dimensions	P_{net}	East - West Direction	
			X	P_{net} X
Operating Load Results	-	232.4 kips	-	1,036 ft-k
Foundation Mat	(0.15 kcf) (14.5')(17')(2.17')	80.2 kips	5 ft	401 ft-k
Soil over Mat	(0.1 kcf)(14.5')(17')(2.5')	61.6 kips	5 ft	308 ft-k
Totals =	-	374.2 kips	-	1,745 ft-k

$X_{na} = \Sigma\ (P_{net}\ X) / \Sigma\ (P_{net}) = (1{,}745\ ft\text{-}k) / (374.2\ kips) = 4.66\ ft$

offset = (distance to mat center) - (X_{na}) = (5 ft) - (4.66 ft) = 0.34 ft

percentage offset = 100 [(0.34 ft) / (14.5 ft)] = 2.3 % < 5 %, OK

SAMPLE DESIGN OF A COMBINED FOUNDATION

Check Overturning Stability

By inspection of the applied loads, Empty plus East to West Seismic controls

Overturning Moment, (in this case, all overturning moments are seismic)

$OTM = \Sigma M + \Sigma Vt + \Sigma P d$ (note that vertical loads cancel out)

$= [2 (246.3 \text{ ft-k}) + (66 \text{ ft-k}) + (98.1 \text{ ft-k}) + (3 \text{ ft-k})] + [2 (14.0 \text{ k}) + (7 \text{ k}) + (10.1 \text{ k}) + (0.2 \text{ k})](2.17 \text{ ft})$

$= 758 \text{ ft-k}$

Resisting Moment, (in this case, all resisting moments are dead weight)

RM = (mat and soil weight)(mat width) /2 + (applied weight)(distance to empty load C.G.)

$= (80.2 \text{ kips} + 61.6 \text{ kips})(14.5 \text{ ft}) /2 + (159.4 \text{ kips})(2.25 \text{ ft} + 4.84 \text{ ft})$

$= 2{,}158 \text{ ft-k}$

Stability Ratio, (for seismic combinations, refer to Practice 670.215.1216)

$SR = 0.85\ RM / OTM = (0.85)(2{,}158 \text{ ft-k}) / (758 \text{ ft-k}) = 2.4 > 1.0$, OK

Note that a condition of one empty vessel plus one full vessel must be investigated if applicable.

Check Sliding Stability

Not required for seismic load combinations. Refer to Practice 670.215.1216.

Check Soil Bearing

area of mat, $A = (14.5 \text{ ft})(17 \text{ ft}) = 247 \text{ sf}$

By inspection of the applied loads, Operating plus East to West Seismic controls

P = (mat and soil weight) + (operating weight) = (80.2 kips + 61.6 kips) + (232.4 kips) = 374.2 kips

V = 2 (22.1 kips) + (9.2 kips) + (13.5 kips) + 0.3 kips) = 67.2 kips

M = Σ (seismic moments) + (V)(mat thickness) + (operating weight)(C.G. offset)

$= [2 (421.7 \text{ ft-k}) + (89.8 \text{ ft-k}) + (133.2 \text{ ft-k}) + (4.5 \text{ ft-k})] + [(67.1 \text{ kips})(2.17 \text{ ft})] + [(232.4 \text{ kips})(0.34 \text{ ft})]$

$= 1{,}296 \text{ ft-k}$

eccentricity,

$e = M / P = (1{,}296 \text{ ft-k}) / (374.2 \text{ kips}) = 3.5 \text{ ft} > b/6$

maximum gross soil bearing,

$SB_{max} = 2 P / 3 a [b/2 - e] = 2 (374.2 \text{ kips}) / 3 (17.0 \text{ ft})[(14.5 \text{ ft}) /2 - (3.5 \text{ ft})] = 3.9 \text{ ksf}$

allowable gross soil bearing,

$SB_{allow} = (SB_{net}) + (\text{soil density})(\text{depth}) = (3.5 \text{ ksf}) + (0.1 \text{ kcf})(2.5 \text{ ft} + 2.17 \text{ ft}) = 4.0 \text{ ksf} > SB_{max}$, OK

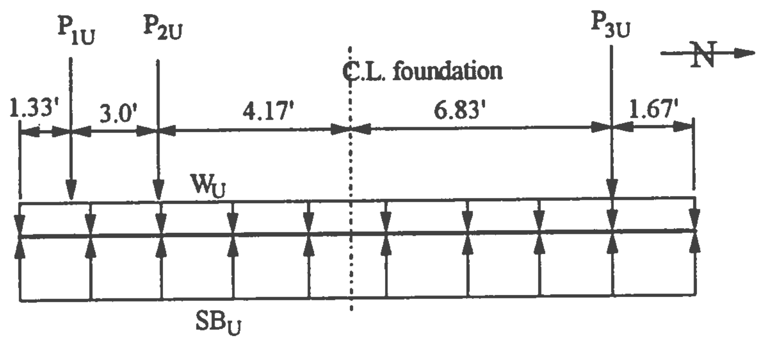
<u>USE 17' 0" by 14' 6" by 2' 2" mat</u>

SAMPLE DESIGN OF A COMBINED FOUNDATION

COMPUTATION OF FORCES FROM FACTORED LOADS

Load cases in the north/south direction are illustrated in this sample calculation. East/west loads, and associated foundation design, must be included in an actual design.

Case #1, Operating



$U = 1.4\ D$

mat & soil weight,
$w_u = 1.4$ (14.5 ft width) [(0.15 kcf concrete)(2.17 ft) + (0.10 kcf soil)(2.5 ft)] = 11.68 klf

sliding end of drum,
$P_{1U} = 1.4\ (P_{net}) = 1.4$ (60 kips) = 84 kips

sliding end of exchanger & pipe support,
$P_{2U} = 1.4\ [\Sigma P_{net}] = 1.4$ [(68 kips) + (3 kips)] = 99.4 kips

anchored end of drum & exchanger,
$P_{3U} = 1.4[\Sigma P_{net}] = 1.4$ [(60 kips) + (46 kips)] = 148.4 kips

$P = \Sigma P_u$ = (11.68 klf)(17 ft) + (84 kips) + (99.4 kips) + (148.4 kips) = 530.4 kips

summed loads on foundation,
$M = \Sigma\ (P_u)(d)$ = (84 kips)(7.17 ft) + (99.4 kips)(4.17 ft) + (148.4 kips)(-6.83 ft) = 3.2 ft-k

eccentricity,
$e = M / P$ = (3.2 ft-k) / (530.4 kips) = 0.006 ft too small to be of significance, neglect

maximum soil bearing,
$SB_{max} = P / L$ = (530.4 kips) / (17 ft) = 31.2 klf

at point #1:

V_{UL} = (31.2 klf - 11.68 klf) (1.33 ft) = 26.0 kips

V_{UR} = (26.0 kips) - (84 kips) = -58.0 kips

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SAMPLE DESIGN OF A COMBINED FOUNDATION

M_U = (26.0 kips)(1.33 ft) /2 = 17.3 ft-k

at point #2:

V_{UL} = (-58.0 kips) + (31.2 klf - 11.68 klf) (3 ft) = -58.0 k + 58.6 k = 0.6 kips

V_{UR} = (0.6 kips) - (99.4 kips) = -98.8 kips

M_U = (31.2 klf - 11.68 klf)(4.33 ft)^2 /2 - (84 kips)(3 ft) = -69.0 ft-k

between point #2 and #3:

V_U = 0 at X = (98.8 kips) / (31.2 klf - 11.68 klf) = 5.06 ft

M_{max} = (31.2 klf - 11.68 klf)(4.33 ft + 5.06 ft)^2 /2 - (84.0 kips)(3 ft + 5.06 ft) - (99.4 kips)(5.06 ft) = -319.4 ft-k

at point #3:

V_{UL} = (-98.8 kips) + (31.2 klf - 11.68 klf)(11 ft) = -98.8 k + 214.7 k = 115.9 kips

V_{UR} = (115.9 kips) - (148.4 kips) = -32.5 kips

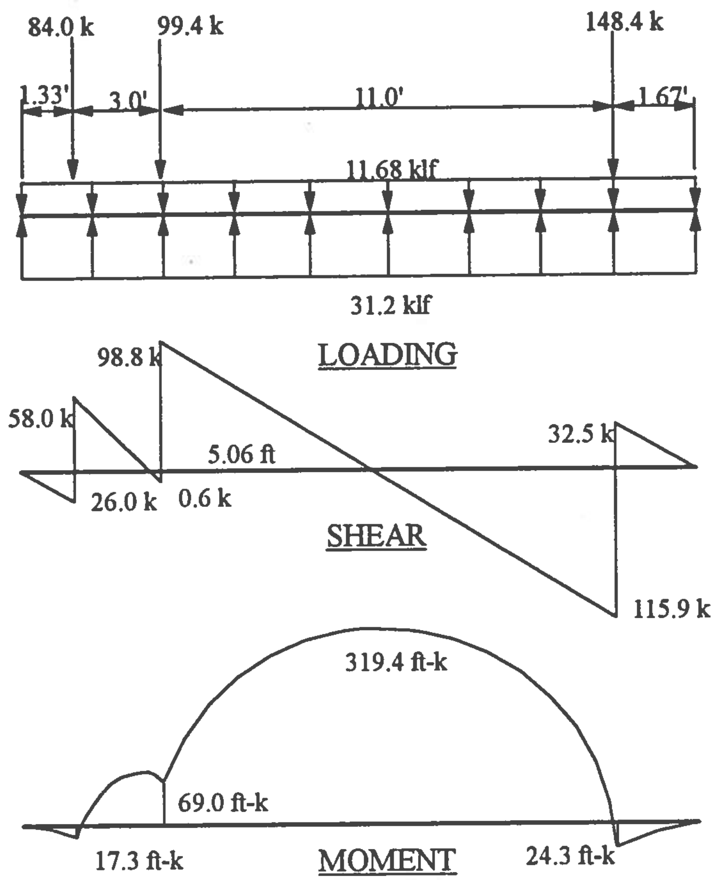
M_u = (31.2 klf - 11.68 klf)(15.33 ft)^2 /2 - (84.0 kips)(14 ft) - (99.4 kips)(11 ft) = 24.3 ft-k

check at right end:

V_u = (-32.5 kips) + (31.2 klf - 11.68 klf)(1.67 ft) = 0.1 kips, close enough

M_u = (24.3 ft-k) - (31.2 klf - 11.68 klf)(1.67 ft)^2 /2 = -2.9 ft-k, close enough

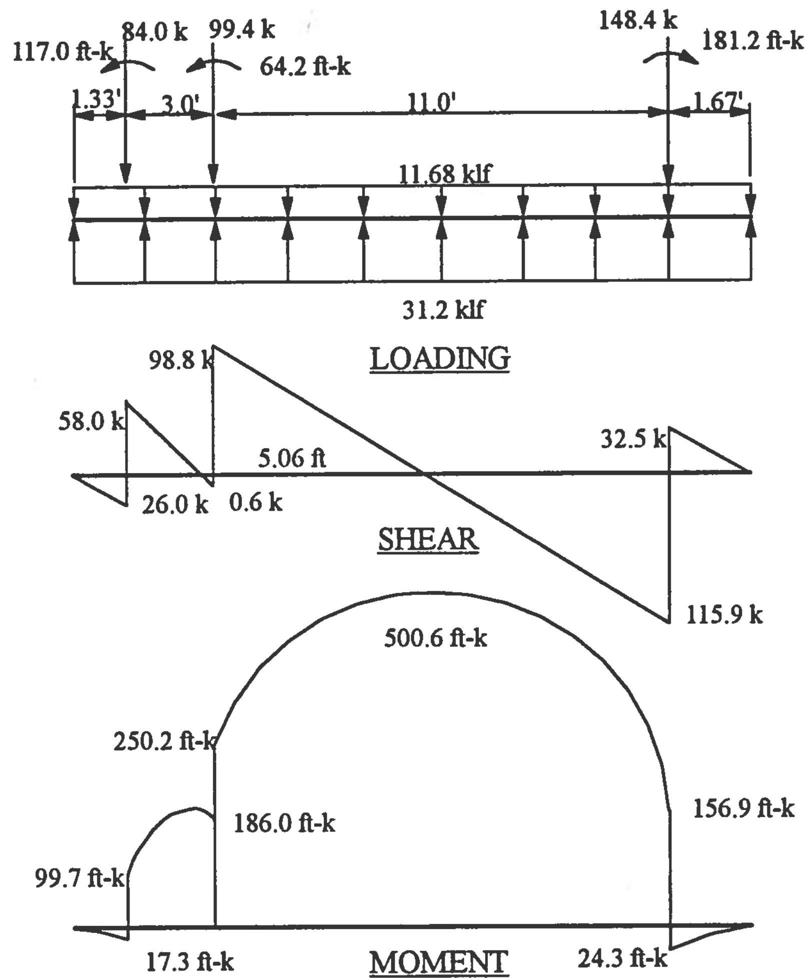
Case #1, Operating:



SAMPLE DESIGN OF A COMBINED FOUNDATION

Case #2, Operating + Friction

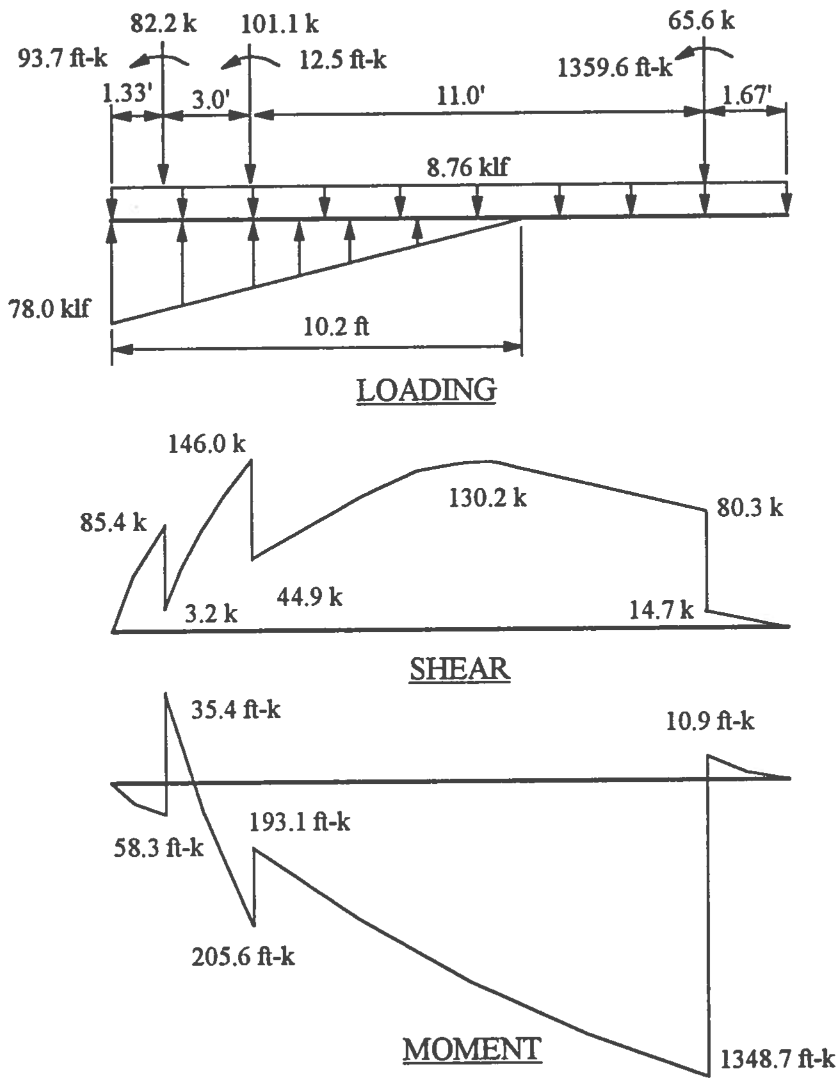
Calculations are omitted for brevity.



SAMPLE DESIGN OF A COMBINED FOUNDATION

Case #3, Operating + Seismic (North to South)

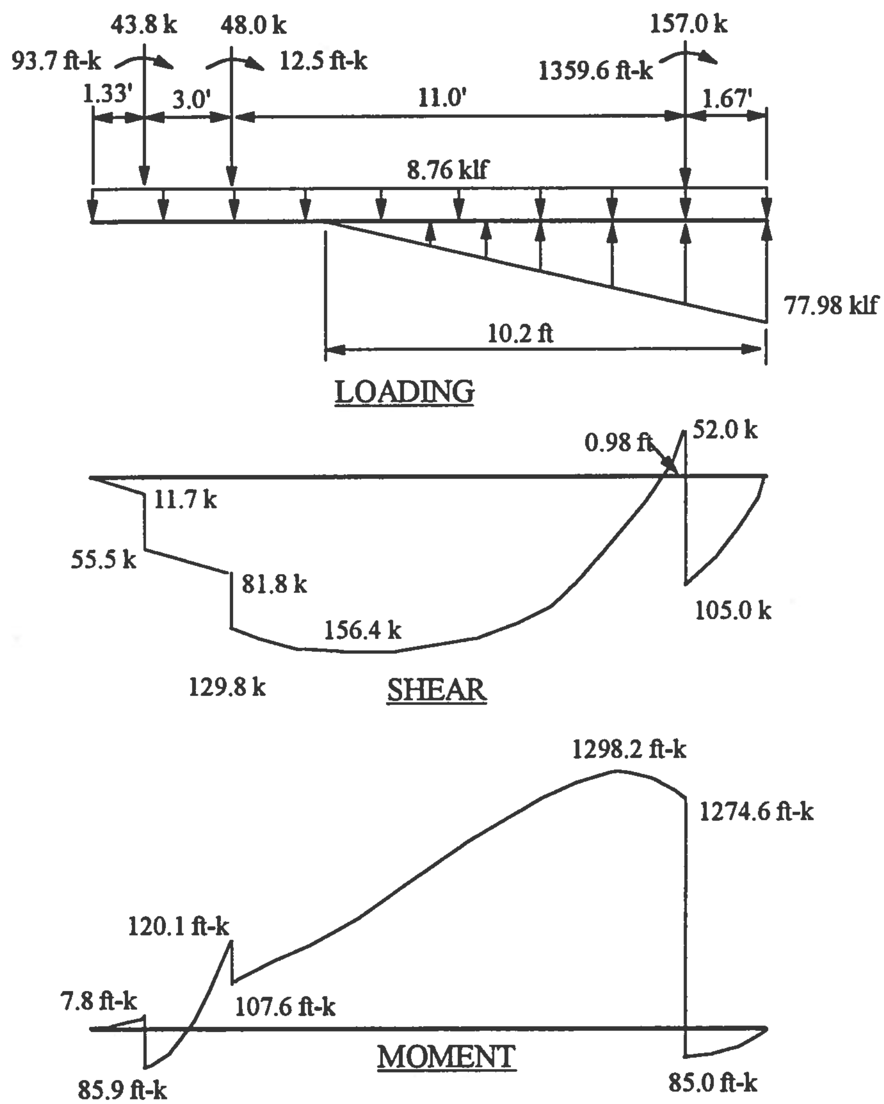
Calculations are omitted for brevity.



SAMPLE DESIGN OF A COMBINED FOUNDATION

Case #4. Operating + Seismic (South to North)

Calculations are omitted for brevity.

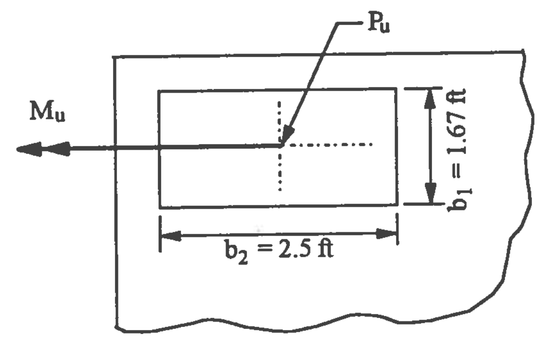


CONCRETE DESIGN

Check Punching Shear (reference: ACI 318-95, and PCA Notes on ACI 318-95)

By inspection, the critical case is at the anchor end of the drum (at the corner of the mat).

Check stress for: Operating & Seismic, North to South



$U = 0.75\ [1.4D + 1.7L + 1.7(1.1)E]$

$P_u = 0.75\ [1.4\ (60\ \text{kips}) + 1.87\ (-13.7\ \text{kips})] = 43.8\ \text{kips}$

$M_u = 0.75(1.87)\ [(39.1\ \text{kips})(2.17\ \text{ft}) + (597.9\ \text{ft-k})] = 957.6\ \text{ft-k}$

$d = (26\ \text{in mat}) - (3\ \text{in clear}) - (\text{say } 0.5\ \text{in bar}) = 22.5\ \text{in}$

from PCA figure 18-16, neglecting overhang of mat,
$b_1 = c_1 + d/2 = (20\ \text{in}) + (22.5\ \text{in})/2 = 31.3\ \text{in}$

$b_2 = c_2 + d/2 = (30\ \text{in}) + (22.5\ \text{in})/2 = 41.3\ \text{in}$

portion of pier moment resisted by mat moment (ACI equation 13-3),
$\gamma_f = 1 / [1 + (2/3)\text{sqrt}\ (b_1 / b_2)] = 1 / [1 + (2/3)\ \text{sqrt}\ (31.3\ \text{in} / 41.3\ \text{in}) = 0.63$

portion of pier moment resisted by mat shear (ACI equation 11-41),
$\gamma_v = 1 - \gamma_f = 1 - (0.63) = 0.37$

section modulus of critical section (PCA figure 18-16),
$J/C = b_1^2\ d\ (b_1 + 4b_2) + d^3\ (b_1 + b_2) / 6b_1$
$= (31.3\text{in})^2\ (22.5\ \text{in})\ [(31.3\ \text{in}) + 4(41.3\ \text{in})] + (22.5\ \text{in})^3\ [(31.3\ \text{in}) + (41.3\ \text{in})] / 6(31.3\ \text{in})$
$= 27{,}468\ \text{in}^3$

area of critical section (PCA figure 18-16),
$A_C = d\ (b_1 + b_2) = (22.5\ \text{in})\ [(31.3\ \text{in}) + (41.3\ \text{in})] = 1634\ \text{in}^2$

pier axial load, minus soil pressure, is resisted as mat shear, (in this case, soil pressure under pedestal is zero)
$V_U = P_U\ (\text{pier}) - (SB_3)(\text{area}) = (43.8\ \text{kips}) - (0\ \text{ksf})(31.3\ \text{in}/12)(41.3\ \text{in}/12) = 43.8\ \text{kips}$

maximum shear stress (PCA page 18-13),
$V_{U1} = V_U / A_C + \gamma_v\ (M_U) / (J/C)$
$= (43.8\ \text{kips}) / (1634\ \text{in}^2) + (0.37)(957.6\ \text{ft-k} * 12) / (27{,}468\ \text{in}^3)$
$= 0.182\ \text{ksi} = 182\ \text{psi}$

determine allowable shear stress from ACI 318-95, section 11.12.2.1,

SAMPLE DESIGN OF A COMBINED FOUNDATION

case (a),
$\beta_C = b_2 / b_1 = (41.3 \text{ in}) / (31.3 \text{ in}) = 1.32$

$V_C = (2 + 4 / \beta_C) \sqrt{f'c}\ bd = (2 + 4 / 1.32) \sqrt{4{,}000 \text{ psi}}\ bd = (318 \text{ psi})\ bd$

case (b),
$\alpha_s = 20$ (corner column)

$b_O = b_1 + b_2 = (31.3 \text{ in}) + (41.3 \text{ in}) = 72.6 \text{ in}$

$V_C = [\alpha_s\, d / b_O + 2] \sqrt{f'c}\ bd = [20\ (22.5 \text{ in}) / (72.6 \text{ in}) + 2] \sqrt{4{,}000 \text{ psi}}\ bd = (519 \text{ psi})\ bd$

case (c),
$V_C = 4 \sqrt{f'c}\ bd = 4 \text{ sqrt } (4{,}000 \text{ psi})\ bd = 4 \sqrt{4{,}000 \text{ psi}}\ bd = (252 \text{ psi})\ bd$

case (c) controls, $v_C = 252$ psi

check applied shear versus nominal,
$V_{U1} = 182 \text{ psi} < V_C = 252 \text{ psi}$, OK

Check Mat Shear Capacity

maximum shear,
$V_U = 156.4$ kips (operating plus seismic, south to north)

d = (26 in mat) - (2 in clear) - (say 0.625 in bar) /2 = 23.69 in

b = (14.5 ft)(12 in/ft) = 174 in

$V_N = \phi \sqrt{f'c}\ bd = (0.85) \sqrt{4{,}000 \text{ psi}}\ (174 \text{ in})(23.69 \text{ in}) / 1000 = 221.6 \text{ kips} > V_U$, OK

Flexural Reinforcing (bottom steel)

maximum moment,
$M_U = 1348.7$ ft-k (operating plus seismic, north to south)

d = (26 in mat) - (3 in clear) - (say 0.625 in bar) /2 = 22.69 in

b = (14.5 ft)(12 in/ft) = 174 in

$K_U = M_U / (bd^2 / 12000) = (1348.7 \text{ ft-k}) / [(174 \text{ in})(22.69 \text{ in})^2 / (12{,}000)] = 181$

from ACI Design Handbook SP-17, for $K_U = 181 \Longrightarrow \rho = 0.0034$

check minimum reinforcing ratio, $\rho_{MIN} = 200 / fy = 0.0033$, OK

$A_s = \rho bd = (0.0034)(174 \text{ in})(22.69 \text{ in}) = 13.4 \text{ in}^2$ (0.92 in^2 per foot) <u>USE #8 bars @ 10 in</u>

SAMPLE DESIGN OF A COMBINED FOUNDATION

Flexural Reinforcing (top steel)

maximum moment,
M_U = 1298.2 ft-k (operating plus seismic, south to north)

d = (26 in mat) - (2 in clear) - (say 0.625 in bar) /2 = 23.69 in

b = (14.5 ft)(12 in/ft) = 174 in

$K_U = M_U / (bd^2 / 12000) = (1298.2 \text{ ft-k}) / [(174 \text{ in})(23.69 \text{ in})^2 / (12{,}000)] = 160$

from ACI design handbook SP-17, $\rho = 0.0030$

check minimum reinforcing ratio, $\rho_{MIN} = 200 / fy = 0.0033$, use ρ_{MIN}

$A_S = \rho_{MIN}\, bd = (0.0033)(174 \text{ in})(23.69 \text{ in}) = 13.6 \text{ in}^2$ (0.94 in² per foot)

USE #8 bars @ 10 in

A similar analysis must be performed for the transverse (East -West) direction